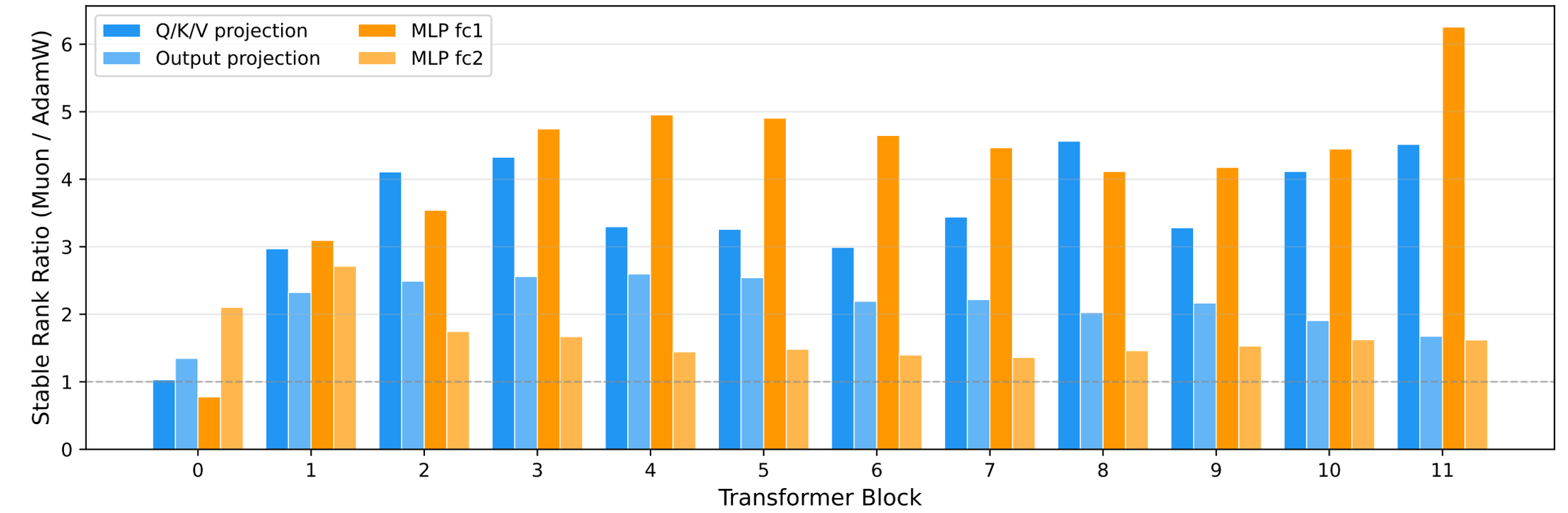
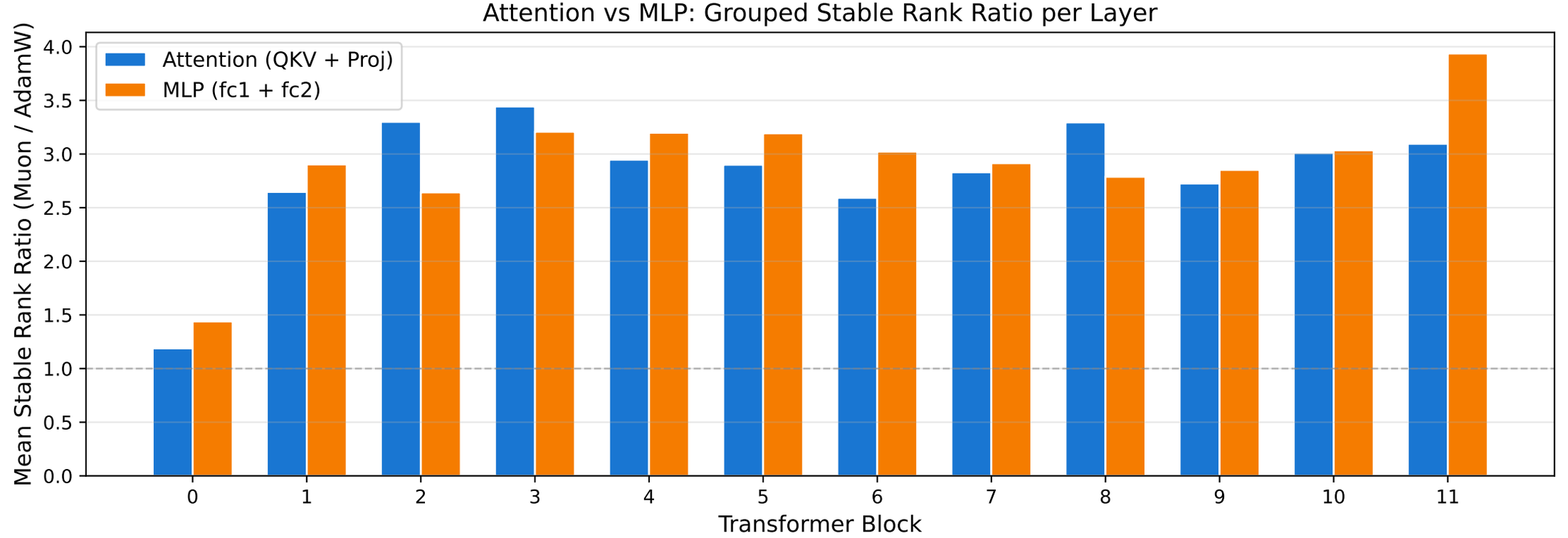

DeiT-S Per-Layer Stable Rank Ratio: Attention vs MLP Components
(Muon 62.26% vs AdamW 50.75%, seed=2, 20 epochs)
Q/K/V projection
Output projection
MLP fc1
MLP fc2
Stable Rank Ratio (Muon / AdamW)
0
1
2
3
4
5
6
0
1
2
3
4
5
6
7
8
9
10
11
Transformer Block


Attention vs MLP: Grouped Stable Rank Ratio per Layer
Attention (QKV + Proj)
MLP (fc1 + fc2)
Mean Stable Rank Ratio (Muon / AdamW)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
1
2
3
4
5
6
7
8
9
10
11
Transformer Block